Phase 3 Report: Intelligence

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Repository: https://github.com/KatarukondaAshok/GrowMar-AI-ML-Project/tree/main/phase-3

1. Task status and evidence

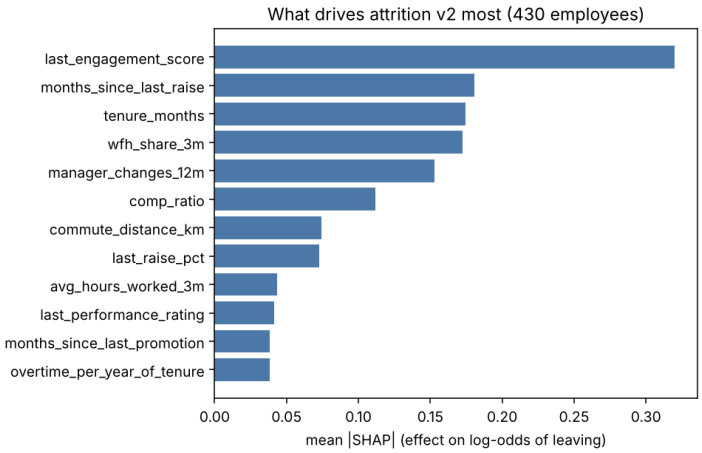
Task	Status	Evidence (path in the repository)
11 Resume ranking	Done	phase-3/Task11_Resume_Ranking/outputs/tables/ranking_quality.csv, misranks_explained.csv
12 Fairness audit	Done	phase-3/Task12_Fairness_Audit/outputs/tables/fairness_summary.csv
13 Attrition v2	Done	phase-3/Task13_Attrition_v2/outputs/tables/comparison_table.csv, tuning_log.csv
14 Risk explanations	Done (read-aloud test pending)	phase-3/Task14_Risk_Explanations/outputs/attrition_risk_2026.csv
15 Pay bands	Done	phase-3/Task15_Pay_Bands/outputs/tables/group_gap_table.csv

2. Proof it works

- **Tests:** `pytest -q` gives **15 passed** (masking in every resume layout, twin resumes, risk bands 43/86/301, plain-English reasons, saved models, pay band inside role band).
- **Task 11:** top-5 precision **0.96**, NDCG@5 0.968, Spearman 0.84 against the human labels (random order: 0.18 / 0.30). No name, gender, age, city or college tier is used.
- **Task 12:** twin gap before masking 0.0562 points (all 25 pairs differ, max 0.378); after masking **0.000** (0 of 25). Ranking quality unchanged (0.96 / 0.968).

Attrition model (5-fold CV, same folds)	ROC-AUC	PR-AUC	Precision top 30%	Recall top 30%
Task 3 baseline: LogReg balanced	0.703	0.301	0.273	0.559
Attrition v2: LightGBM	0.725	0.339	0.284	0.581

- **Task 13:** per-fold gain +0.022 on average but spread 0.036 and 3 of 5 folds won, so the gain is **not proven**. Calibration: Brier 0.114 vs 0.125 base rate; calibration error 0.028 to 0.020. No leakage (strongest single feature AUC 0.61).
- **Task 14:** 430 rows in the official format; bands 43 High / 86 Medium / 301 Low; three plain-English reasons each; every row says it needs human review.
- **Task 15:** pay error (MAE) Rs 54,293 vs Rs 115,964 for the role-average baseline; 80.7% of true offers fall inside the recommended range. A model that sees career gap and gender copies the past gap (-2.5 points); the final job-only model does not.



3. What I learned

- Fairness is measured, not assumed: even without gender, names and tier words moved scores until I masked them.
- A fix must be tested in every layout: cutting the first lines of a resume missed names in two-column and table layouts.
- Compare models fold by fold: a +0.02 gain with 0.04 spread is not a proven gain.
- Explanations show drivers, not certainty, and must be worded so they do not judge the person (tenure, education).

4. Problems I hit and how I solved them

Problem	Fix
Masking by position left twin gaps in 4 pairs	Removed name, city and e-mail by value; added a two-column test
First risk reasons printed model-scaled numbers (e.g. -1.5/5)	Used the person's real values; missing values leave the number out
Pay range covered only 43% of offers	Clipping to the 2026 band hid older, lower offers; evaluate without clipping (80.7%), clip only new offers
Starter helpers (cortexhr_helpers.py) not supplied	Wrote group_gap() and risk_bands() with tests
v2 is not clearly better than the baseline	Reported it honestly; ship for calibration and explainability with the baseline as fallback

5. Plan for the next phase

- Phase 4: forecasting and putting everything live in the same repository (folder phase-4).
- To add before submitting: read 10 risk rows aloud to a friend and fix confusing wording (Task 14).